



Computer Basic

INTRODUCTION TO COMPUTER

A computer is an electronic device, operating under the control of instructions stored in its own memory that can accept data (input), process the data according to specified rules, produce information (output), and store the information for future use.



FUNCTIONALITIES OF A COMPUTER

Any digital computer carries out **five functions** in gross terms.



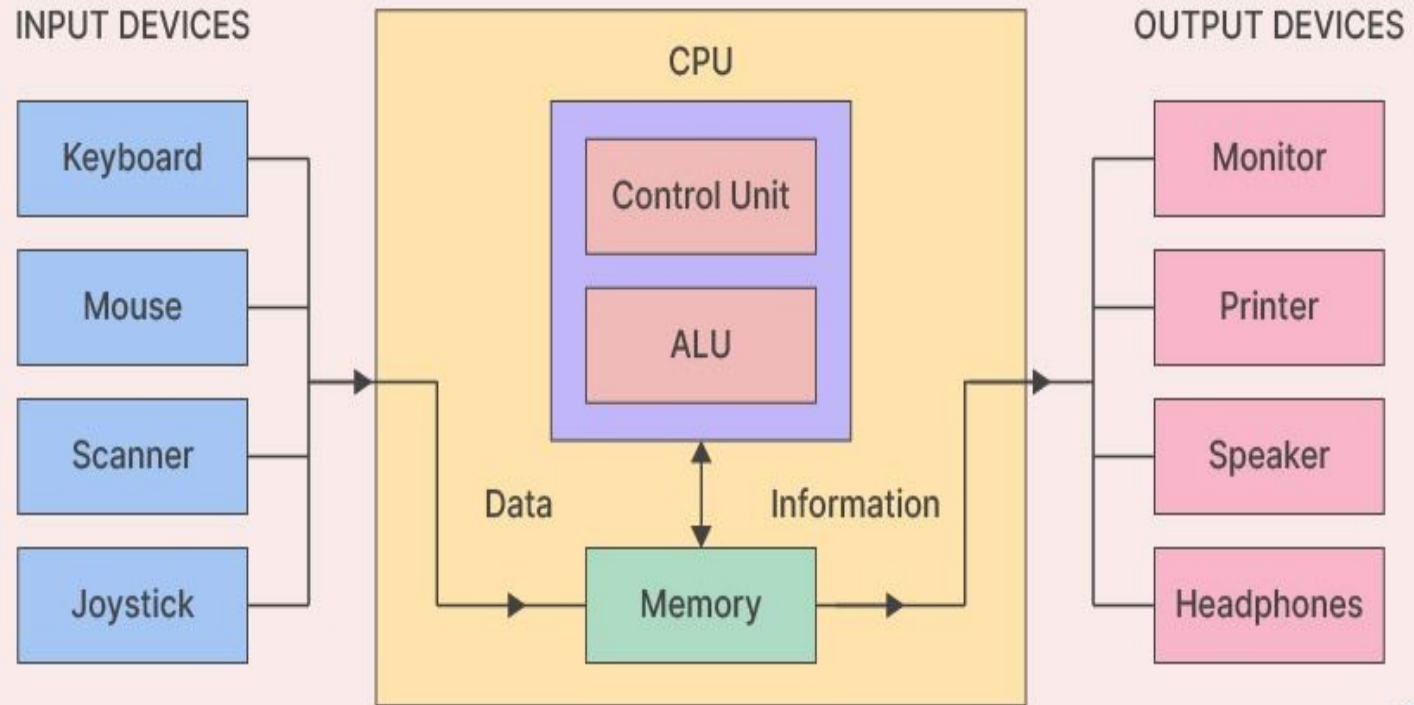
- ❑ Takes Data as Input
- ❑ Stores the data/Instructions in its memory and use them when required.
- ❑ Processes the data and converts it into useful Information.
- ❑ Generates the Output.
- ❑ Control all the above four steps.





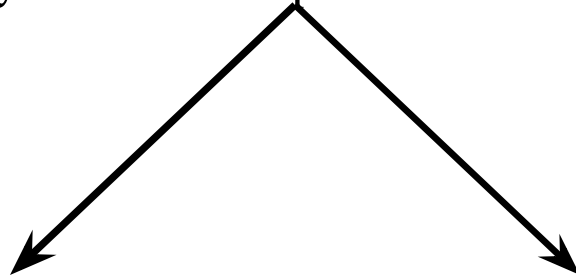
COMPUTER BASIC ARCHITECTURE

Basic Architecture of a Computer



COMPUTER COMPONENTS

Any kind of computers consists of



HARDWARE

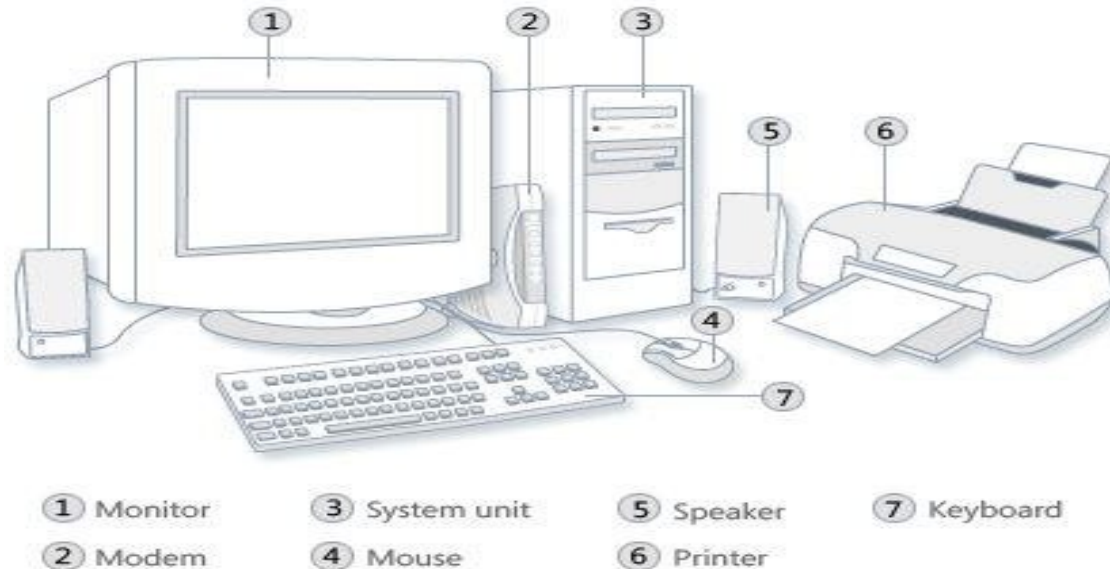
SOFTWARE.



HARDWARE

Computer hardware is the collection of physical elements that constitutes a computer system.

Computer hardware refers to the physical parts or components of a computer such as the monitor, mouse, keyboard, computer data storage, hard drive disk (HDD), system unit (graphic cards, sound cards, memory, motherboard and chips), etc. all of which are physical objects that can be touched.



INPUT DEVICES

Input device is any peripheral (piece of computer hardware equipment to provide data and control signals to an information processing system such as a computer or other information appliance.

Input device Translate data from **form** that humans understand to one that the computer can work with. Most common are keyboard and mouse

Examples of Manual Input Devices			
Keyboard 	Numeric Keypad 	Pointing Device 	Remote Control 
Joystick 	Touch Screen 	Scanner 	Graphics Tablet 
Microphone 	Digital Camera 	Webcams 	Light Pens 



EXAMPLE OF INPUT DEVICES:-

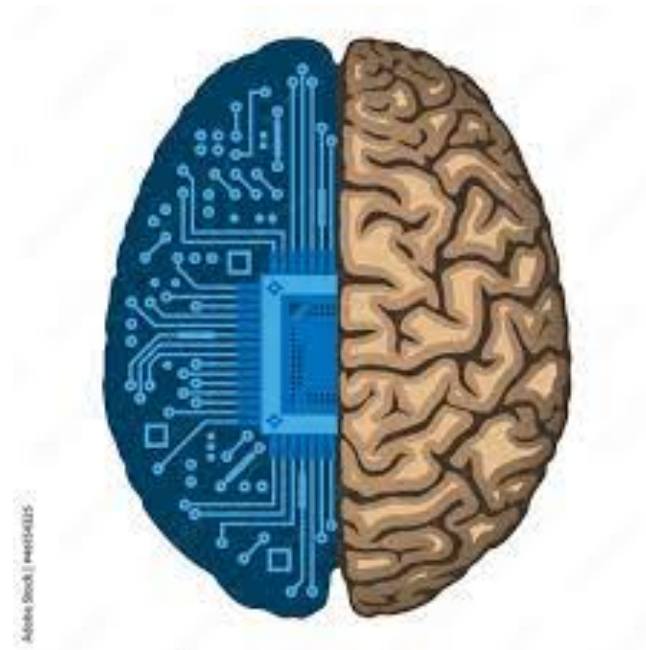
1. Keyboard	2. Mouse (pointing device)	3. Microphone
4. Touch screen	5. Scanner	6. Webcam
7. Touchpads	8. MIDI keyboard	
10.Graphics Tablets	11.Cameras	12.Pen Input
13.Video Capture Hardware	14.Microphone	15.Trackballs
16.Barcode reader	17.Digital camera	18.Joystick
19.Gamepad	20.Electronic Whiteboard	

Note: The most common use keyboard is the QWERTY keyboard. Generally standard Keyboard has 104 keys.



CENTRAL PROCESSING UNIT (CPU)

A CPU is brain of a computer. It is responsible for all functions and processes. Regarding computing power, the CPU is the most important element of a computer system.





CENTRAL PROCESSING UNIT (CPU)

The CPU is comprised of three main parts :

1. Arithmetic Logic Unit (ALU): Executes all arithmetic and logical operations. Arithmetic calculations like as addition, subtraction, multiplication and division. Logical operation like compare numbers, letters, or special characters

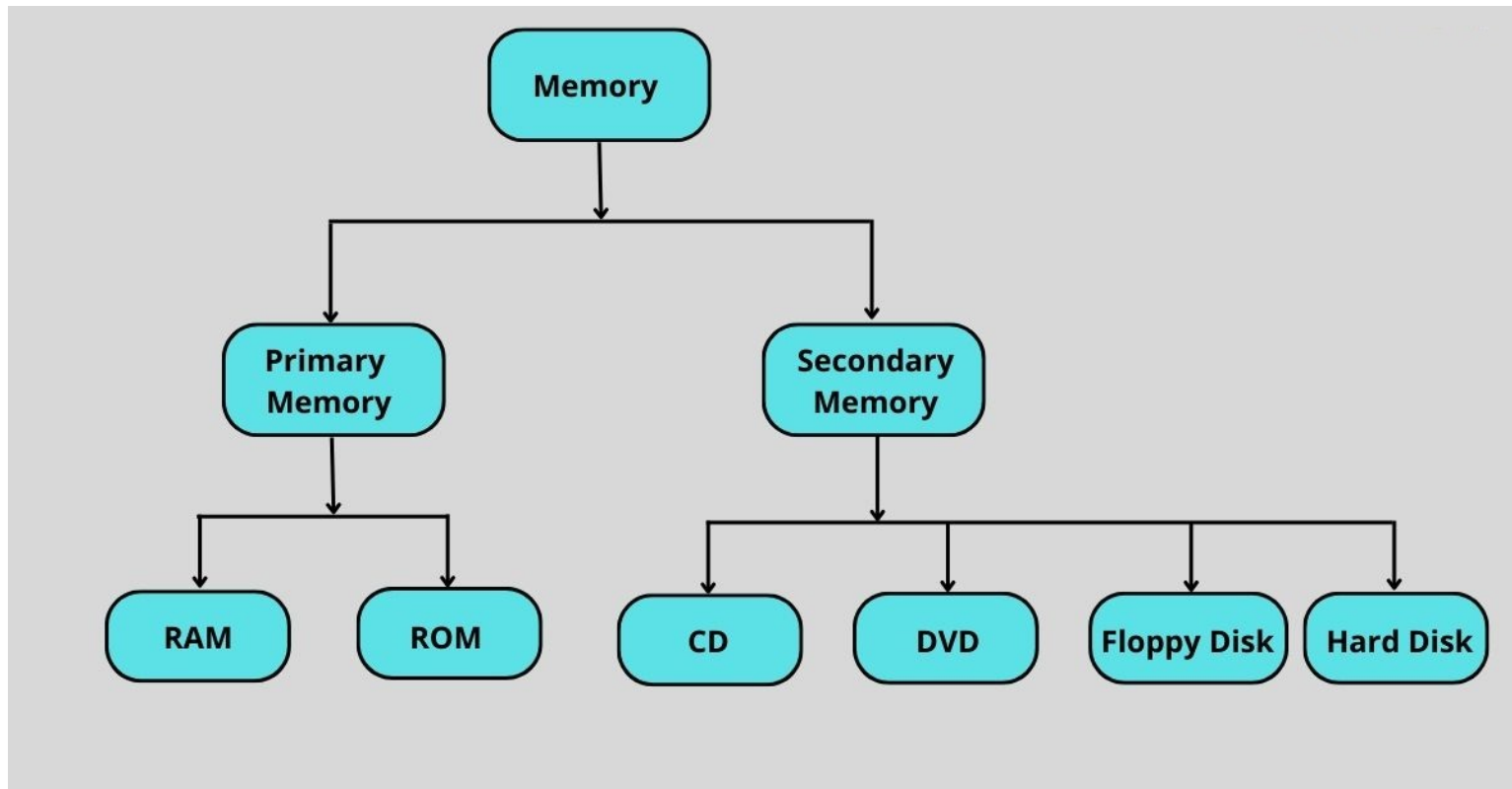
2. Control Unit (CU): controls and co-ordinates computer components.

- ❑ Read the code for the next instruction to be executed.
- ❑ Provide the necessary data to an ALU or register.
- ❑ Instruct to do necessary actions

3. Registers :Stores the data that is to be executed next, "very fast storage area".



COMPUTER MEMORY





PRIMARY MEMORY

- ❑ **RAM:** Random Access Memory (RAM) is a memory scheme within the computer system responsible for storing data on a temporary basis. it can be accessed by the processor when needed.

It is **volatile** in nature, which means that data will be erased once supply to the storage device is turned off.

RAM stores data randomly and the processor accesses these data randomly from the RAM storage. RAM is considered "random access" because you can access any memory cell directly.

- ❑ **ROM (Read Only Memory):** ROM stands for Read-Only Memory. It is a non-volatile memory (permanent memory) that is used to store important information which is used to operate the system. As its name refers to read-only memory, we can only read the programs and data stored on it.



SECONDARY MEMORY

Stores data and programs permanently: its retained after the power is turned off

Hard drive (HD): A hard disk is part of a unit, often called a "disk drive," "hard drive," or "hard disk drive," that store and provides relatively quick access to large amounts of data on an electromagnetically charged surface or set of surfaces.

Optical Disk: an optical disc drive (ODD) is a disk drive that uses laser light as part of the process of reading or writing data to or from optical discs.

Compact discs (CD), DVDs, and Blu-ray discs are common types of optical media.



OUTPUT DEVICES

An output device is any piece of computer hardware equipment used to communicate the results of data processing carried out by an information processing system (such as a computer) which converts the electronically generated information into human-readable form.

Examples of Output Devices			
CRT Monitor 	TFT Monitor 	Laser Printer 	Inkjet Printer 
Dot Matrix Printer 		Speakers 	
Plotters 		Multimedia Projectors 	



COMPUTER SOFTWARE

Software is a generic term for organized collections of computer data and instructions, often broken into two major categories:

- ❑ **System Software** is necessary to manage the computer resources and support the execution of application programs. Software like operating systems, compilers, editors and drivers etc., come under this category.

A computer cannot function without the presence of these.

- ❑ **Applications Software** – Application software, or app for short, is software that performs **specific tasks** for an end-user. For example, Microsoft Word or Excel are application software, as are common web browsers such as Firefox or Google Chrome.



COMPARISON APPLICATION SOFTWARE AND SYSTEM SOFTWARE

	System Software	Application Software
Example:	Microsoft Windows Linux Unix Mac OSX DOS	Opera (Web Browser) Microsoft Word (Word Processing) Microsoft Excel (Spreadsheet software) MySQL (Database Software) Microsoft PowerPoint (Presentation Software) Adobe Photoshop (Graphics Software)
Interaction:	Generally, users do not interact with system software as it works in the background.	Users always interact with application software while doing different activities.
Dependency:	System software can run independently of the application software.	Application software cannot run without the presence of the system software.

UNIT OF MEASUREMENTS

Storage measurements: The basic unit used in computer data storage is called a bit (binary digit).

Computers use these little bits, which are composed of ones and zeros, to do things and talk to other computers. All your files, for instance, are kept in the computer as binary files and translated into words and pictures by the software (which is also ones and zeros).

This two number system, is called a “binary number system” since it has only two numbers in it. The decimal number system in contrast has ten unique digits, zero through nine.





COMPUTER ARCHITECTURE & ORGANIZATION

ARCHITECTURE & ORGANIZATION

Architecture is those attributes visible to the programmer

- **Architecture** deals with the design of computers, data storage devices, and networking components that store and run programs, transmit data, and drive interactions between computers, across networks, and with users.
- Computer architects use parallelism and various strategies for memory organization to design computing systems with very high performance.

Organization is how features are implemented

- Computer Organization is how operational parts of a computer system are linked together. It implements the provided computer architecture.
- Control signals, interfaces, memory technology, Circuit design. It deals with low-level design issues.

ARCHITECTURE & ORGANIZATION

CT Difference Between Computer Architecture and Computer Organization

Computer Architecture	Computer Organization
Architecture describes what the computer does.	The Organization describes how it is done.
Computer Architecture deals with the functional behavior of computer systems.	Computer Organization deals with the structural relationship.
For designing a computer, its architecture is fixed first.	For designing a computer, an organization is decided after its architecture.
As a programmer, you can view architecture as a series of instructions, addressing modes, and registers.	The implementation of the architecture is called organization.
Architecture focuses on its hardware.	Whereas Organization focuses on its performance.



THANK YOU